

SIM1002 Calculus I
Tutorial 11
Natural Logarithms

Exercise 7.2

1. $\ln(t - 2) = \ln 8 - \ln t$. Solve for t .

In Exercises 2-7, find the derivative of y with respect to x , t , or θ , as appropriate.

2. $y = t\sqrt{\ln t}$

3. $y = \frac{1+\ln t}{t}$

4. $y = \ln(\ln(\ln x))$

5. $y = \sqrt{\ln \sqrt{t}}$

6. $y = \ln \sqrt{\frac{(x+1)^5}{(x+2)^{20}}}$

7. $y = \int_{\sqrt{x}}^{\sqrt[3]{x}} \ln t \, dt$

Evaluating Integrals.

Evaluate integrals in Exercises 8-11.

8. $\int_2^4 \frac{dx}{x \ln x}$

9. $\int_2^{16} \frac{dx}{(2x\sqrt{\ln x})}$

10. $\int_0^{\frac{\pi}{12}} 6 \tan 3x \, dx$

11. $\int \frac{\sec x dx}{\sqrt{\ln(\sec x + \tan x)}}$

Logarithmic Differentiation

In Exercises 12-15, use logarithmic differentiation to find the derivative of y with respect to the given independent variable.

12. $y = \sqrt{\frac{1}{t(t+1)}}$

13. $y = \tan \theta \sqrt{2\theta + 1}$

$$14. \quad y = \sqrt{\frac{(x+1)^{10}}{(2x+1)^5}}$$

$$15. \quad y = \sqrt[3]{\frac{x(x+1)(x+2)}{(x^2+1)(2x+3)}}$$

Exercises 7.3

Solving Exponential Equations

In Exercises 1-2, solve for t .

$$1. \quad e^{x^2} e^{2x+1} = e^t$$

$$2. \quad e^{-2t} + 6 = 5e^{-t}$$

Finding Derivatives

In Exercises 3-6, find the derivative of y with respect to x , t , or θ , as appropriate.

$$3. \quad y = e^{4\sqrt{x}+x^2}$$

$$4. \quad y = \ln(3\theta e^{-\theta})$$

$$5. \quad y = \ln\left(\frac{\sqrt{\theta}}{1+\sqrt{\theta}}\right)$$

$$6. \quad y = \int_{e^{4\sqrt{x}}}^{e^{2x}} \ln t \, dt$$

In Exercises 7-8, find $\frac{dy}{dx}$.

$$7. \quad \tan y = e^x + \ln x$$

$$8. \quad \ln y = xe^y - 2$$

Finding Integrals

Evaluate the integrals in Exercises 9-12.

$$9. \quad \int \frac{e^{-\sqrt{r}}}{\sqrt{r}} dr$$

$$10. \quad \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} (1 + e^{\cot \theta}) \operatorname{cosec}^2 \theta \, d\theta$$

$$11. \quad \int_0^{\sqrt{\ln \pi}} 2xe^{x^2} \cos(e^{x^2}) \, dx$$

12. $\int \frac{dx}{1+e^x}$

Exercises 7.5

Use L'Hopital's rule to find the limits in Exercises 1-5

1. $\lim_{\theta \rightarrow \frac{\pi}{4}} \frac{\tan \theta - 1}{\theta - \frac{\pi}{4}}$

2. $\lim_{x \rightarrow 0^+} \frac{\ln(e^x - 1)}{\ln x}$

3. $\lim_{y \rightarrow 0} \frac{\sqrt{ay + a^2} - a}{y}$

4. $\lim_{x \rightarrow 0^+} (\ln x - \ln \sin x)$

5. $\lim_{h \rightarrow 0} \frac{e^h - (1+h)}{h^2}$

Indeterminate Powers and Products

Find the limits in Exercises 6-8.

6. $\lim_{x \rightarrow e^+} x^{\frac{1}{x-1}}$

7. $\lim_{x \rightarrow 0} (e^x + x)^{\frac{1}{x}}$

8. $\lim_{x \rightarrow 0^+} x(\ln x)^2$